

Object Detection Metrics & Post-processing

IoU, NMS, and mAP

Goals

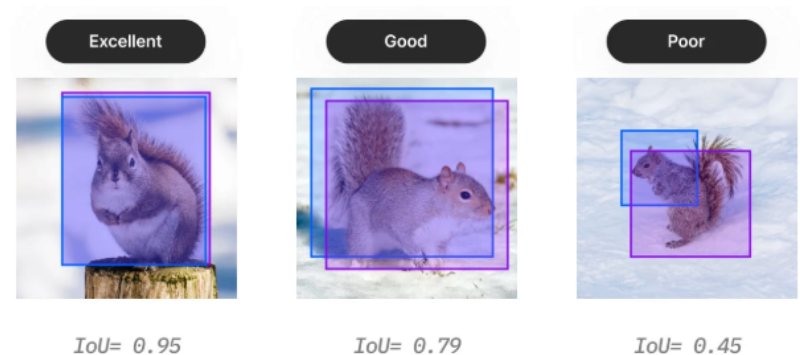
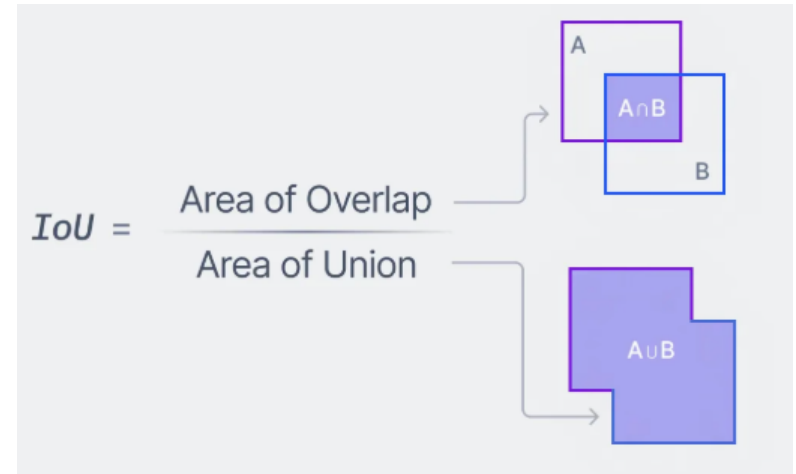
- Understand how detections match to ground truth via IoU
- Know how overlapping predictions are pruned with NMS
- Interpret precision, recall, AP, and mAP; avoid pitfalls

Detector Outputs Refresher

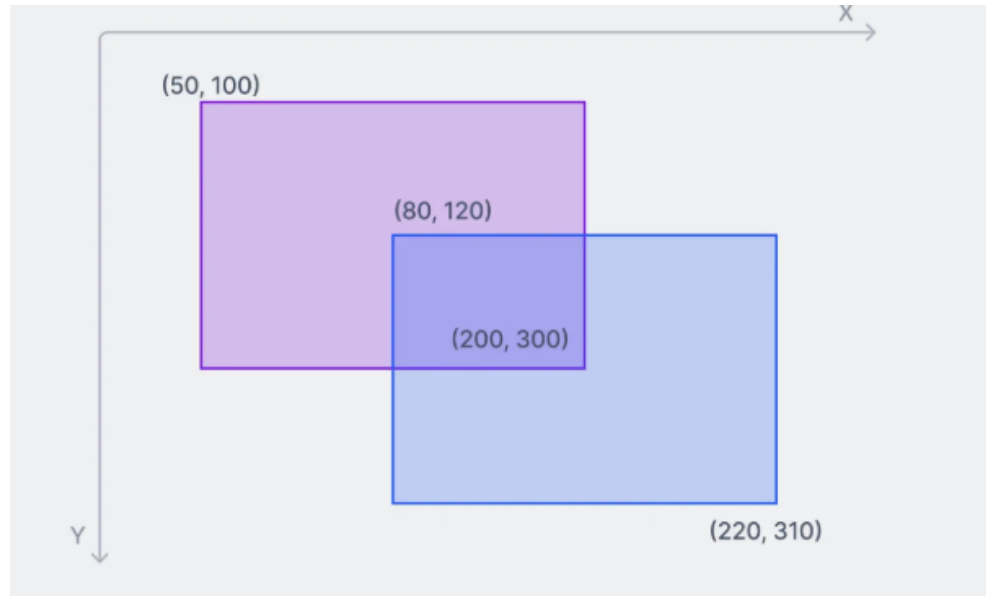
- For each image returns: boxes (x1,y1,x2,y2), class scores, confidence/objectness
- Multiple detections may target the same object → need post-processing
- Evaluation ties predictions to ground truth using IoU + class match

Intersection over Union (IoU)

- Measures quality between predicted and ground-truth boxes
- Range [0,1]; higher is better
- Used for matching: **TP** if class is correct **AND** $\text{IoU} \geq \text{threshold}$ (e.g., 0.5)



IoU example



Intersection = $(200 - 80) \times (300 - 120) = 21,600$

Purple = $(200 - 50) \times (300 - 100) = 30,000$

Blue = $(220 - 80) \times (310 - 120) = 26,600$

$\text{IoU} = 21,600 / 56,600 = 0.62$

Non-Max Suppression (NMS) — Why and How

- Why: detectors output many overlapping boxes per object
- Greedy NMS: sort by score, keep best, drop overlaps above IoU_{th}
- Variants:
 - Per-class vs class-agnostic

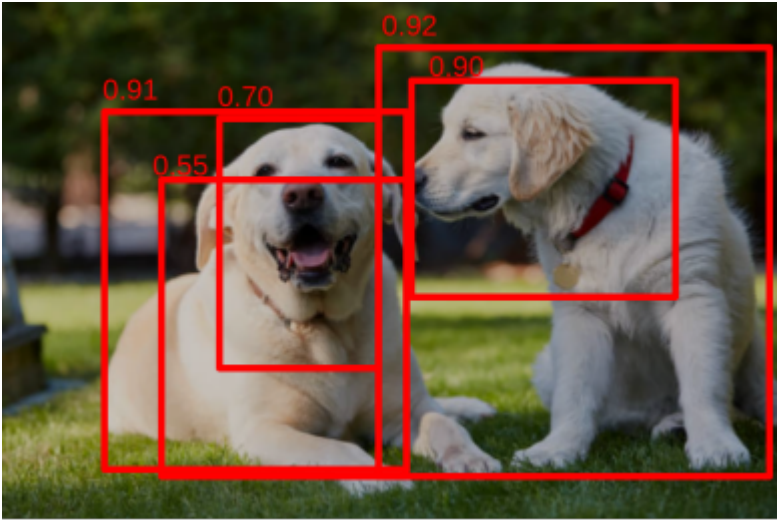
NMS example



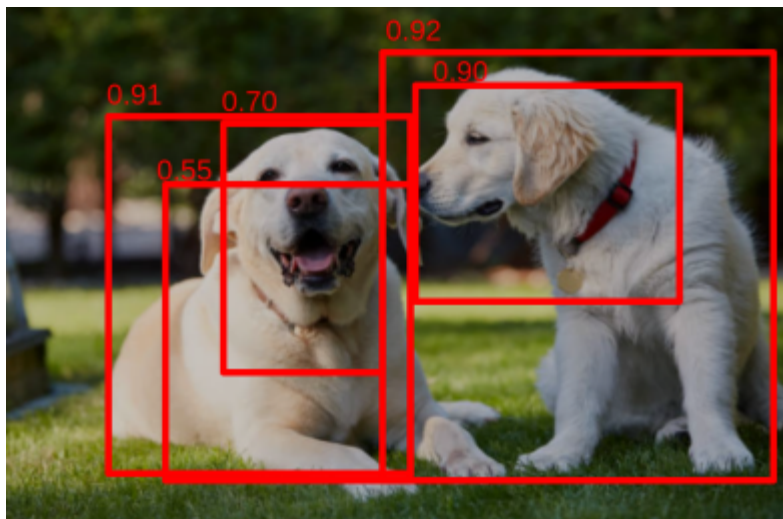
For this Image

NMS example

Object detector proposed these bounding boxes



NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

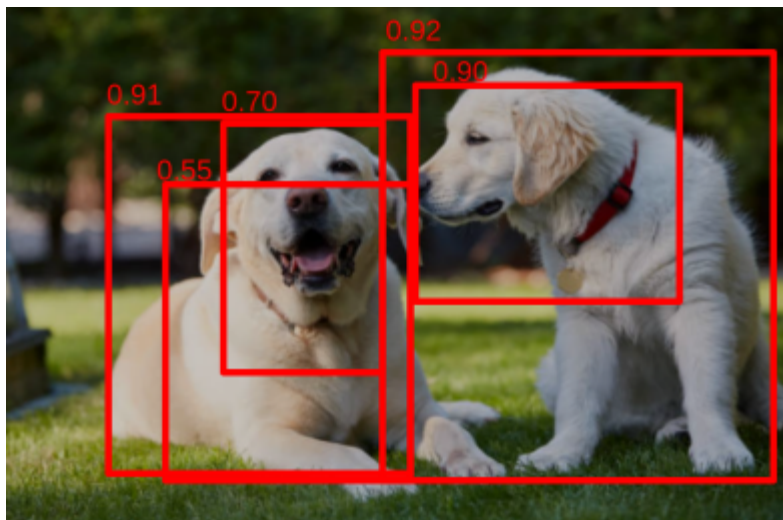
Proposals

Box #	Score	IOU
1	.91	
2	.70	
3	.55	
4	.92	
5	.90	

Keepers

Box #	Score

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

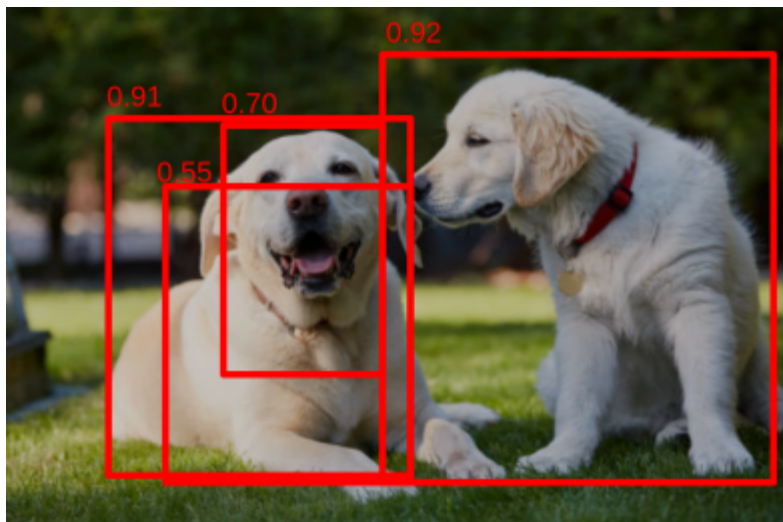
Proposals

Box #	Score	IOU
1	.91	0
2	.70	0
3	.55	0
5	.90	0.3

Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

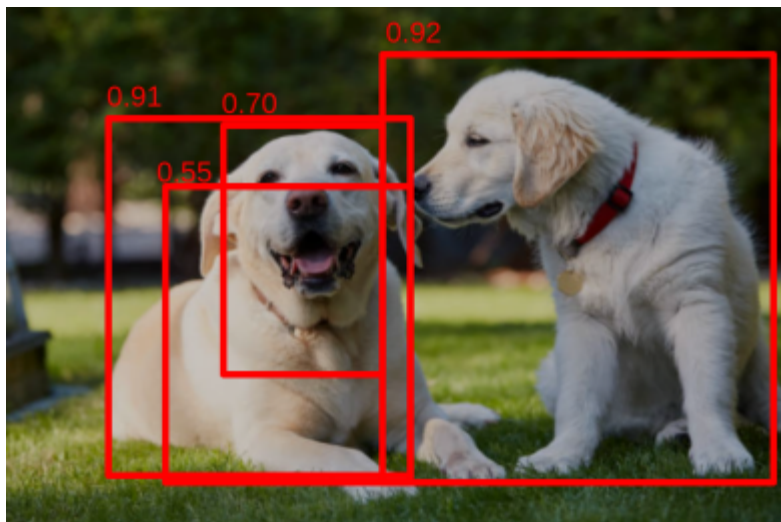
Proposals

Box #	Score	IOU
1	.91	0
2	.70	0
3	.55	0

Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

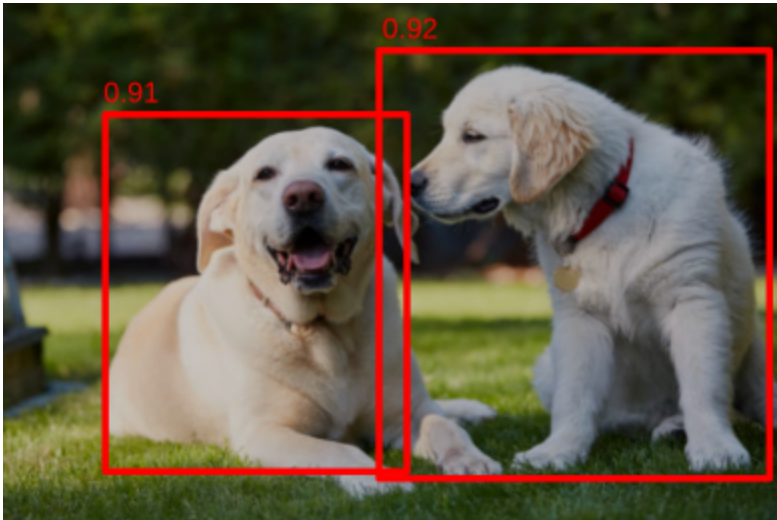
Proposals

Box #	Score	IOU
2	.70	.3
3	.55	.7

Keepers

Box #	Score
4	.92
1	.91

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

Proposals

Box # Score IOU

Keepers

Box # Score

4	.92
1	.91